

UNIVERSITY OF CLAUDE · FREE RESOURCE MAP

Resource Map

CSCE 637 Complexity Theory — module-by-module mapping onto free university courseware

HOW TO USE THIS

This is your lecture hall. For each module: watch or read the mapped external material first, then work the module slides and notes, which are written to assume you have seen it.

Where a specific lecture number is given, that is the one that covers this module. Where a book chapter is given, read that chapter, not the whole book.

Everything here is free. Resources remain under their own licences — link to them, do not redistribute them.

Primary sources for the whole course

Resource	Role in this course
Arora & Barak — Computational Complexity: A Modern Approach (free draft)	The primary source, and the full draft is free from the authors. Chapters 1–11 and 17–20 are this course, and the barrier results of Modules 07 and 09 are treated better here than anywhere else.
Sipser — Introduction to the Theory of Computation, chapters 7–10	The gentler path through Modules 01–07 , and the same book as CSCE 627 — so the two courses share a text, which makes the computability/complexity boundary easy to follow.
MIT 18.404J — the complexity half (free lectures and problem sets)	Free video lectures covering Modules 01–07 , by Sipser, with graded problem sets and solutions.
Garey & Johnson — Computers and Intractability	The catalogue of NP-complete problems , and still the reference for Module 04's reductions. Its appendix of 300 problems is what you reduce from in practice. Library copy.
Cygan et al. — Parameterized Algorithms (free PDF)	Module 12's reference, free in full — the most practically useful material in the course, and the least commonly taught.
Aaronson — P =? NP (free survey)	The honest survey of the central question , including the barriers of Modules 07 and 09 and an assessment of what the evidence is actually worth. Read it twice.

Module-by-module map

Module 01 — Measuring Cost

Resource	What it covers for this module
Arora & Barak — chapter 1 (free draft)	The machine model, the robustness argument of §1, and the definition of a complexity class.
Sipser — chapter 7, sections 7.1–7.2	The same material more gently , with the encoding and measurement issues of §2 stated carefully.
Spielman & Teng — Smoothed Analysis of Algorithms (free)	§3's third row , and the resolution of the simplex puzzle — a good example of theory catching up with practice.
Lipton & Regan — Galactic algorithms (free blog series)	§4's third bullet , where the term comes from, with examples.

Module 02 — P, and What Tractable Means

Resource	What it covers for this module
Arora & Barak — chapter 1, and the discussion of the thesis (free draft)	P, the thesis, and the objections of §1 , stated carefully.
Edmonds — Paths, Trees, and Flowers (free)	§2's callout, in the original — and the introduction is where the thesis is argued. Worth reading for the argument rather than the algorithm.
Babai — Graph Isomorphism in Quasipolynomial Time (free)	§2's fourth row — the current state of the best-known NP-intermediate candidate.
Agrawal, Kayal & Saxena — PRIMES is in P (free)	§2's second row , and remarkably short for what it settled.

Module 03 — NP, and Verification

Resource	What it covers for this module
Arora & Barak — chapter 2 (free draft)	Both definitions of NP, their equivalence, and the structural material of §4.
Cook — The Complexity of Theorem-Proving Procedures (free)	The 1971 paper , which is short and is where the question originates.
Aaronson — P =? NP (free survey)	§2 and §3 assessed honestly — the consequences, the evidence, and what the evidence is actually worth. The best thing written on the question for a general technical audience.
Ladner — On the Structure of Polynomial Time Reducibility (free)	§4's theorem , with the alternating construction.

Module 04 — NP-Completeness

Resource	What it covers for this module
Arora & Barak — chapter 2, and Sipser chapter 7 (free draft)	Cook–Levin with the tableau construction , and the standard reductions.
Garey & Johnson — Computers and Intractability	§2's library, in full — three hundred problems with their status, and the guide to constructing reductions in chapter 3. Library copy, and still the reference.
Karp — Reducibility Among Combinatorial Problems (free)	The 1972 paper that produced the first twenty-one , and turned Cook's single theorem into a field.
Levin — Universal sequential search problems (translation, free)	The independent 1973 formulation , which is two pages and frames the question differently.

Module 05 — Beyond NP

Resource	What it covers for this module
Arora & Barak — chapters 4–5 (free draft)	co-NP, PSPACE, and the polynomial hierarchy — the reference for this module.
Sipser — chapter 8	PSPACE, TQBF, and the games , with the generalised geography reduction worked in full.
Hearn & Demaine — Games, Puzzles, and Computation	§2's games, systematically — which puzzles are NP-complete, PSPACE-complete, or EXPTIME-complete, and why. Library copy.
Beame & Pitassi — Propositional Proof Complexity (free survey)	§4's first argument — why short refutations are not expected, which is the main line of attack on NP versus co-NP.

Module 06 — Space

Resource	What it covers for this module
Arora & Barak — chapter 4 (free draft)	Space classes, Savitch, and Immerman–Szelepcsényi , with both proofs.
Sipser — chapter 8	The same material more slowly , and the input-tape convention of §1 explained carefully.
Reingold — Undirected Connectivity in Log-Space (free)	§1's 2004 result , which is a genuine recent success and uses expander graphs in a surprising way.
Greenlaw, Hoover & Ruzzo — Limits to Parallel Computation (free)	§4's catalogue of P-complete problems, free in full — the reference to check before parallelising anything.

Module 07 — Hierarchies and Barriers

Resource	What it covers for this module
Arora & Barak — chapters 3 and 23 (free draft)	The hierarchy theorems and all three barriers — chapter 23 is the best treatment of the barriers available.
Baker, Gill & Solovay — Relativizations of the $P = ? NP$ Question (free)	§2's result , and both oracle constructions.
Aaronson & Wigderson — Algebrization: A New Barrier in Complexity Theory (free)	§2's third barrier , with a good account of why it was needed.
Aaronson — $P = ? NP$, the barriers section (free)	All three barriers assessed together , with a candid view of what remains.

Module 08 — Randomised Complexity

Resource	What it covers for this module
Arora & Barak — chapters 7 and 20 (free draft)	The randomised classes, and the derandomisation chapter — the reference for §1 and §3.
Nisan & Wigderson — Hardness vs Randomness (free)	§3's central result , and the title is the whole point.
Impagliazzo & Kabanets — Derandomizing Polynomial Identity Tests Means Proving Circuit Lower Bounds (free)	§3's converse direction , which ties the two questions together.
Motwani & Raghavan — Randomized Algorithms	The algorithm-design companion , and CSCE 658's main text. Library copy.

Module 09 — Circuits

Resource	What it covers for this module
Arora & Barak — chapters 6, 14, and 23 (free draft)	Circuits, the known lower bounds, and the natural proofs barrier — chapter 23 covers §3 properly.
Razborov & Rudich — Natural Proofs (free)	§3's barrier, in the original — and the definitions of constructive and large are worth reading exactly.
Håstad — Computational Limitations of Small-Depth Circuits	§2's switching lemma and the parity lower bound , free from the author.
Aaronson — $P = ? NP$, the circuit sections (free)	§2's calibration and §4's assessment , including the 5n figure and what it means.

Module 10 — Interactive Proofs

Resource	What it covers for this module
Arora & Barak — chapters 8 and 11 (free draft)	Interactive proofs, $IP = PSPACE$, and the PCP theorem — the reference for this module.
Shamir — $IP = PSPACE$ (free)	§2's theorem , with the arithmetisation worked through.
Arora, Lund, Motwani, Sudan & Szegedy — Proof Verification and the Hardness of Approximation Problems (free)	§3's theorem , and the paper that created the hardness-of-approximation field.
Goldwasser, Micali & Rackoff — The Knowledge Complexity of Interactive Proof Systems (free)	§1 and §4's originating paper — the model and zero knowledge, in one place.

Module 11 — Hardness of Approximation

Resource	What it covers for this module
Williamson & Shmoys — The Design of Approximation Algorithms (free PDF)	§1's table from the algorithm side , free in full, and the same book as CSCE 669 Module 10.
Arora & Barak — chapters 11 and 22 (free draft)	PCP's consequences and the gap technique of §1 , with the reductions.
Khot — On the Unique Games Conjecture (free survey)	§2, by the person who proposed it — including an honest account of the evidence against.
Raghavendra — Optimal Algorithms and Inapproximability Results for Every CSP? (free)	§2's striking consequence — the generic SDP being optimal under UGC.

Module 12 — Fine-Grained and Parameterised

Resource	What it covers for this module
Cygan et al. — Parameterized Algorithms (free PDF)	Free in full, and the reference for §1 and §2 — the most practically useful book in this course.
Williams — On Some Fine-Grained Questions in Algorithms and Complexity (free survey)	§3's hypotheses and their consequences , surveyed by one of the field's founders.
Bringmann — Fine-Grained Complexity Theory (free lecture notes)	§3 taught , with the edit-distance lower bound worked through.
Downey & Fellows — Fundamentals of Parameterized Complexity	The founders' treatment , including the W-hierarchy of parameterised intractability that this module does not cover. Library copy.

Module 13 — What Complexity Tells You

Resource	What it covers for this module
Aaronson — $P \stackrel{?}{=} NP$ (free survey)	§1 and §2, done properly by someone in the field — and the best single thing to read after this course.
Arora & Barak — the whole book, reread (free draft)	Free, and the second reading is different from the first — particularly chapters 23 and 20.
Cygan et al. — Parameterized Algorithms (free PDF)	If one book from this course should stay on your desk, it is this one — Module 12 is where the practical value is.
Garey & Johnson, appendix; and the modern successors	The reduction library , which is what you will actually use — keep a copy or a bookmark.